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# A perspective on impedance matching and resonance absorption mechanism for electromagnetic wave absorbing

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ABSTRACT

Broadband absorption is urgently desirable for microwave stealth or photothermal conversion, but is a significant challenge in thin layer applications. Impedance mismatch and undesired dielectric relaxation are the main obstacles to realizing simultaneous resonant absorption at multiple frequency bands. In this perspective, the key issues in broadband absorption are briefly reviewed, which must be addressed by the community to enable microwave absorption promoting and widening. Then, impedance matching coefficient used to identify microwave absorbing ability is highlighted. Finally, an ideal model of dielectric dispersion for carbon-based materials is proposed to be used to design broadband absorbing materials, after deducing the inner nature of the typical quarter-wave resonance absorption existing in absorbers.

## 1. Introduction

The continuous advancements in 5G wireless communication technology have catapulted the world into a smart era, with a large number of wireless devices in widespread use. However, a mass of electromagnetic pollution has been produced, and the undesired microwave pollution always causes serious electromagnetic interference among electronic components, electromagnetic information leakage, and is harmful to human health [1–6]. As a result, the pursuit of efficient microwave absorbers that can effectively dissipate microwave into Joule heat has been regarded as a feasible solution [7–11]. On the other hand, efficient light-absorbing materials are also urgently needed in the fields of solar thermal utilization and seawater desalination. There are many types of electromagnetic wave (EMW) absorbing materials, including traditional EMW absorbing materials represented by ferrites [12–14] and magnetic metals [15,16], and new types of EMW absorbing materials represented by carbon materials [17–19], MXene [20–22], topological insulator [23,24], conducting polymer matrix composites [25,26] and metal sulphides [27,28]. These materials can boost EMW absorption and solar thermal conversion, yet suffer from poor absorbing bandwidth in thin layer applications.

Obtaining wider bandwidth and stronger absorption at thin thickness is always a hot pursuit in the research community. In recent years, considerable progress has been made in the field of high-performance wideband absorbent materials [29–34]. A lot of strong absorption materials ($RL_{min}<-60$ dB) have been reported. For example, Xie et al. [35] developed a hierarchical aerogel incorporating liquid metal-MXene components that exhibits excellent EMW absorption capability with a remarkable attenuation of up to −73.2 dB. Wu et al. [36] prepared a heterobimetallic disulfide nanoparticles composite dispersed in $CoS_2$ hollow sea urchins by wet impregnation method and triple calcination, which exhibits excellent EMW absorption performance at a thickness of 1.97 mm with an $RL_{min}$ of −75.23 dB. Che et al. [37] fabricated MOF-derived $ZnFe_2O_4$–ZnO–Fe@C microspheres with custom-built heterogeneous interfaces via a synergistic confined growth strategy. The magnetic particles reduced by pyrolysis treatment can promote the graphitization transition from carbon-containing organic ligands, and exhibit excellent EMW absorption with a thickness of only 2.0 mm and an $RL_{min}$ of −66.5 dB. Compared to increasing absorption strength, it is more challenging to achieve broadband absorption at thin thickness. The absorption bandwidth can be widened by optimizing the impedance matching through the design of hollow structure, heterogeneous interface and other microstructures. For example, Ji et al. [38] prepared Co/$Co(OH)_2$@PCN composites via a collaborative strategy of

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lyophilization and hard templates, which achieved an effective absorption bandwidth of up to 7.1 GHz at 2.2 mm by controlling the carbonization temperature. Kong et al. [39] prepared hierarchical $Co_2P/CoS_2$@C@$MoS_2$ composites with hollow cavities and multiple hetero-interfaces via a hydrothermal method, which reached an effective absorption bandwidth of 8.96 GHz at a thickness of 4 mm. Duan et al. [40] prepared carbonyl iron (CI)/carbon foam (CF) magnetic-dielectric composites by carbonization and vacuum impregnation. Benefiting from the synergistic regulation of dielectric and magnetic losses, CI/CF composites have an EAB of 8.48 GHz at 1 mm thickness. Lu et al. [41] proposed the PC@CuS composites with the effective absorption bandwidth up to 7.8 GHz at 1.85 mm thickness by optimizing the impedance matching of the composites. Most efforts on broadband absorption in thin layer applications rely on multi-resonance coexistence. The strategy requires a deep understanding on quarter-wavelength resonant absorption mechanism. Greater progress on coating absorber can hardly achieve without the support of an ideal theoretical modeling.

A perfect absorbing material is one with relative complex permeability same as the relative complex permittivity, so that its wave impedance is equal to that of air. But this material is impossible to exist in nature. Quarter-wavelength resonance is the most fundamental way to achieve strong EMW absorption in thin absorbers [42,43], but it is obtained only in a narrow frequency range suffering from the thickness sensitivity caused by the dielectric dispersion relation. A deeper understanding of quarter-wavelength resonance is urgently needed to produce multi-frequency resonance for designing broadband and strong absorption materials. Impedance matching design has always been an important means to achieve wide-band electromagnetic wave absorption, especially for multilayer or impedance gradient absorbers. The EMW absorbing materials with good impedance matching between the absorber and air can allow more EMW to enter into the materials, so as to achieve the purpose of absorbing EMWs.

## 2. Impedance matching theory

For EMW absorbing layer on the metal backplate, there are two interfacial reflections when the EMW is incident on the absorber. One is the interface between air and the absorbing material, and the other is the interface between the absorbing material and the metal plate (Fig. 1a). The metal plate has a high electrical conductivity and is able to completely reflect incident EMWs. In this case, the EMW absorption performance of the material is generally evaluated by the reflection loss (*RL*), which is described as follows [44].

$$RL = 20\ log\left|\frac{Z_{in}/Z_0 - 1}{Z_{in}/Z_0 + 1}\right| \tag{1}$$

$$Z_{in} = Z_0\sqrt{\mu_r/\varepsilon_r}\,\mathrm{Tanh}\left[j(2\pi fd/c)\sqrt{\mu_r\varepsilon_r}\right] \tag{2}$$

Here $Z_{in}$ is the input impedance of an absorber, $Z_0$ is wave impedance in free space (120π Ω), $d$ is the absorber thickness, $\varepsilon_r$ is the relative complex permittivity ($\varepsilon_r = \varepsilon'$-j$\varepsilon''$), $\mu_r$ is the relative complex permeability ($\mu_r = \mu'$-j$\mu''$), $f$ and $c$ are frequency and optical speed in free space, respectively.

When the sum of the reflected wave energy of the two interfaces is equal to the incident wave energy and their phases are opposite, the EMW will be completely absorbed. In this case, the input impedance should be equal to wave impedance in free space. However, the input impedance of the absorber is usually a complex number, so it is not equal to 120π Ω in most cases. Constructing an impedance matching parameter that can accurately evaluate the absorbing property of materials is crucial for designing broadband EMW absorbers.

In order to evaluate the impedance matching degree of EMW absorbing materials, the input impedance matching coefficient is described by Eq. (3) in some works [45].

$$Z_m = |Z_{in}/Z_0| \tag{3}$$

When $Z_{in} = Z_0$, i.e., $Z_m = 1$, and then it is an input impedance matching state. It is also expected that the closer the input impedance matching coefficient is to 1, the better the impedance matching of the EMW absorber. However, for a microwave absorber with a metal backplate (Fig. 1a), it is misused and not suitable in most cases. Fig. 2a and 2b shows contour maps far the $Z_m$ and *RL* for any input impedance, respectively. It is obviously observed that the two are not one-to-one monotonic interrelated. For example, the values of $Z_m$ at points A and B in Fig. 2a are both 1. According to Eq. (3), both points A and B should be in impedance matching state. However, only point A in Fig. 2b is impedance matched and point B is impedance mismatched, which indicates that Eq. (3) is only applicable in some cases and not in most cases. Therefore, we proposed a new formula of impedance matching coefficient as follows [46].

$$M_z = \frac{2Z_{in}'/Z_0}{|Z_{in}/Z_0|^2 + 1} \tag{4}$$

Here $Z_{in}'$ means the real input impedance. It is evident from Fig. 2b and 2c that there is a one-to-one correspondence between *RL* and $M_z$ for any input impedance. When $M_z = 1$ (point A in Fig. 2c), the ideal impedance matching is achieved, indicating that the EMW absorption performance is the best at this time. The closer $M_z$ is to 1, the better the impedance matching of the EMW absorber will be. The effective absorption region in Fig. 2b is consistent with that of $M_z > 0.82$ shown in Fig. 2c. The wider the range of $M_z > 0.82$, the wider the effective absorption bandwidth of the EMW absorbing materials can be achieved.

For an EMW absorber with infinite thickness or thickness much larger than the wavelength (Fig. 1b), there is only one interfacial reflection when the EMW is incident on the absorber. A good EMW absorbing material can not only rarely reflect the electromagnetic wave, but also completely absorb the incident energy in the shortest possible distance. The absorption performance is determined by the wave impedance matching coefficient ($M_\eta$) and the attenuation constant ($\alpha$). They can be calculated by the following equations, respectively [47,48].

$$M_\eta = \frac{2\eta'}{|\eta|^2 + 1} \tag{5}$$

$$\alpha = \frac{\sqrt{2}\pi f}{c}\sqrt{(\mu''\varepsilon'' - \mu'\varepsilon') + \sqrt{(\mu''\varepsilon'' - \mu'\varepsilon')^2 + (\mu'\varepsilon'' + \mu''\varepsilon')^2}} \tag{6}$$

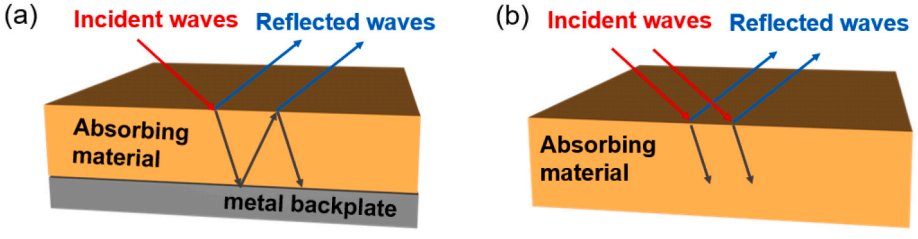



**Fig. 1.** Schematic of EMWs propagation for an absorber (a) with a metal backplate and (b) without a metal backplate. (A colour version of this figure can be viewed online.)

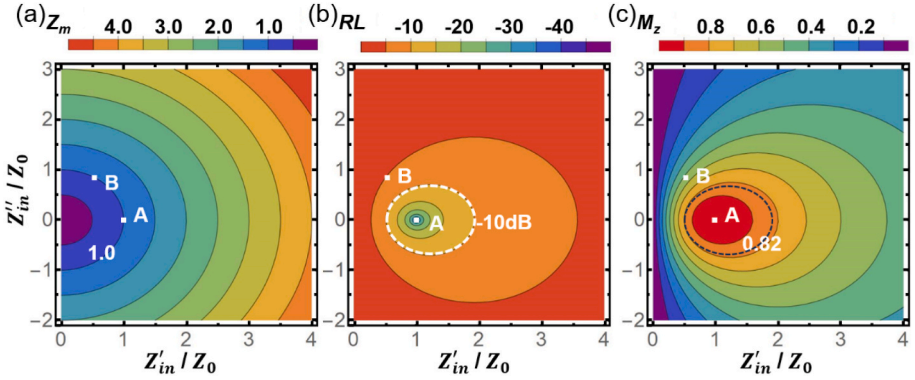



**Fig. 2.** Contour maps of (a) $Z_m$, (b) *RL* and (c) $M_z$ with any input impedance. (A colour version of this figure can be viewed online.)

Here relative wave impedance $\eta = \sqrt{\mu_r/\varepsilon_r}$, $\eta'$ is the real part of relative wave impedance. When the value of $M_\eta$ is closer to 1, it indicates that more EMW enter into the absorber. The larger the α value is, the more thoroughly the electromagnetic wave attenuates inside the absorber. An absorbing material with better performance requires both large $M_\eta$ and α.

## 3. Quarter-wavelength resonance and dielectric parameter dispersion

One common attribute standouts among all successful EMW absorbing coating materials reported so far is that the absorption is dominated by quarter-wavelength resonance. This implies that increasing absorption broadband requires compatible inherent resonances of multiple frequencies. Then, it is crucial to deeply understand the resonance mechanisms of the electromagnetic response. Fig. 3a shows the electrical length dependence of *RL* of carbon materials with different dielectric loss tangent. It is observed that the resonant positions of all three materials are deviated from a quarter-wavelength. The dielectric loss tangents (*Tan δ*) of three materials are 0.4, 0.5, and 0.6, and the corresponding permittivity is 10, 6, and 4, respectively. This implies that the shift of the resonance peak position is related to the loss tangent. We assume that the offset is *Δ*, so the quarter-wavelength model can be given by

$$d = \frac{(2m+1)c}{nf}\left(\frac{1}{4}+\Delta\right), (m = 1, 2, 3\ldots) \tag{7}$$

Here *d* is the thickness of an absorber, *f* is the resonant frequency, *m* is a natural integer, and *n* is the refractive index ($n = Re[\sqrt{\varepsilon_r\mu_r}\,]$). From Eq. (1), Eq. (2), and Eq. (7), Δ can be expressed as

$$\Delta = \frac{hSinh(t\pi)}{4\pi(1 - htCosh(t\pi))} \tag{8}$$

Here $h = \frac{(t - Tan\ \delta_m)}{(1 + tTan\ \delta_m)}$, $t = \frac{k}{n}$, and k is extinction coefficient. $Tan\ \delta_m$ is magnetic loss tangent. From Eq. (8), it can be seen that when $t = Tan\ \delta_m$, i.e., the $Tan\ \delta$ is equal to $Tan\ \delta_m$, $\Delta = 0$. When $t > Tan\ \delta_m$, there will be $\Delta > 0$. When $t < Tan\ \delta_m$, there will be $\Delta < 0$. Fig. 3b and 3c shows the distribution of the maximum electric field for non-magnetic materials and magnetic materials by CST simulation, respectively. It is observed that the resonance thickness of the material dominated by dielectric loss is greater than λ/4, while that of the material dominated by magnetic loss is less than λ/4. This phenomenon is also confirmed by standing waves in the external electric field of the material (Fig. 3e and 3f). The introduction of Δ in the resonant thickness allows to obtain a standing wave ratio (VSWR) equal to 1, when the reflection coefficient is equal to 0. For an EMW absorber plate without the metal backplate, there are also two interfacial reflections when the EMW is incident on the absorber. One is the interface between air and the absorbing material,

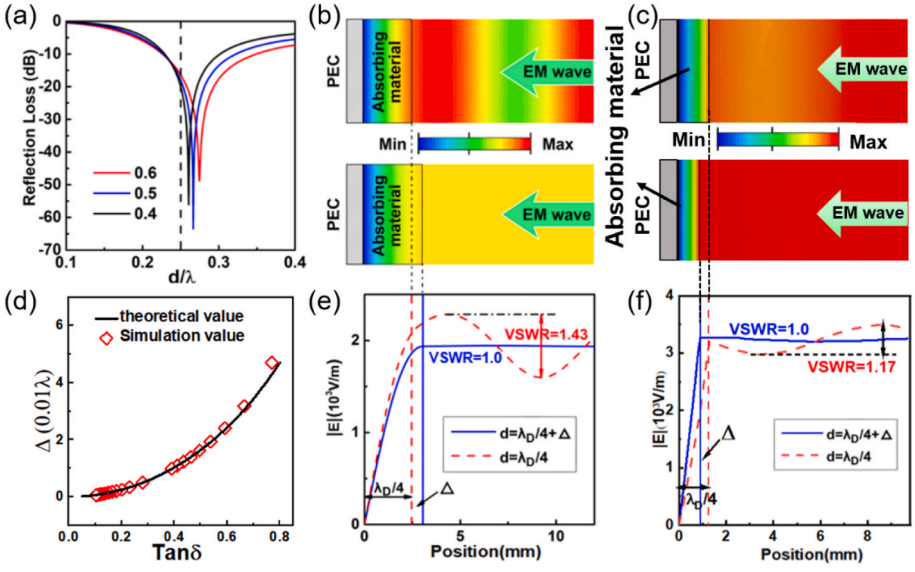



**Fig. 3.** (a) Electrical length dependence of *RL* with different Tan*δ*. Distribution of the maximum electric field for (b) non-magnetic and (c) magnetic absorber. (d) Dependence of Δ on Tan*δ*. Voltage stand waves for (e) non-magnetic and (f) magnetic absorber. (A colour version of this figure can be viewed online.)

and the other is the interface between the absorbing material and air. Its absorption mechanism is the half-wavelength resonance [49,50], which is also subject to an offset, but different from the quarter-wavelength resonance. For an EMW absorber with a high loss material as a backplate, two interfacial reflections are similar to that with a metal backplate. The quarter-wavelength theory is also applicable if the metal backplate is replaced by a high loss material.

For carbon materials, $\mu_r = 1$, and $h = t$, Δ can be further simplified as follows.

$$\Delta = \frac{tSinh(t\pi)}{4\pi(1 - t^2\ Cosh(t\pi))} \tag{9}$$

According to $n - ik = \sqrt{\varepsilon_r \mu_r}(\mu_r = 1)$, we have

$$t = \frac{\sqrt{1 + Tan^2\ \delta} - 1}{Tan\ \delta}, \tag{10}$$

$$n = \sqrt{\varepsilon_r^{'}}\sqrt{\frac{\sqrt{1 + Tan^2\ \delta} + 1}{2}} \tag{11}$$

According to Eq. (9) and Eq. (10), Δ can be simplified as

$$\Delta = \frac{Tan^2\ \delta}{16} \tag{12}$$

Eq. (12) can be used to predict the Δ value when $Tan\ \delta < 0.6$. Based on Eq. (12), the relationship between Δ and $Tan\ \delta$ is plotted in Fig. 3d, which presents the corrections for the quarter-wavelength resonance for non-magnetic materials.

According to Eq. (2), Eq. (7), Eq. (11), and Eq. (12), we have

$$\varepsilon_r^{'} = \frac{c^2}{8f^2d^2}\frac{1 + Tan^2\ \delta/2}{1 + \sqrt{1 + Tan^2\ \delta}} \tag{13}$$

$$\varepsilon_r^{''} = \frac{c}{\pi f d} \tag{14}$$

For carbon materials, the dielectric dispersion relation of ideal absorber can be determined by Eq. (13) and Eq. (14), which is the theoretical support for realizing multi-frequency resonance and can be used to guide the preparation of broadband EMW absorbing materials.

These two formulas can also be used for the design of light-absorbing materials in photothermal conversion applications. Fig. 4 shows the frequency dispersion of $\varepsilon_r^{'}$, $\varepsilon_r^{''}$ and Tanδ of ideal absorbing materials with different thickness. The typical values for the dielectric dispersion relation are shown in Table 1. It is obvious that both $\varepsilon_r^{'}$ and $\varepsilon_r^{''}$ show a rapid decreasing trend with the frequency increases. The corresponding *RL* and $M_z$ are shown in Fig. 5, all the $M_z$ are greater than 0.999, indicating good impedance matching, and the corresponding *RL* are less than −30 dB, indicating good microwave absorption. These indicate that this dielectric dispersion relation can be used to design microwave absorber. The reason why the absorbing performance at high frequency is worse than that at low frequency is that the error of Eq. (12) becomes larger due to the larger loss tangent at high frequency. Up to date, no natural material has been found to well meet this dispersion relation. Consequently, the design of a new dielectric material that possesses such relaxation would be highly pursued in the near future. The large gradient dispersion permittivity for perfect absorption remains a challenge, maybe with more opportunities in developing novel metamaterials.

Some works have been reported on the construction of carbon-based EMW absorbing materials through the design of dielectric parameters. For example, Song et al. [51] proposed a universal permittivity-attenuation evaluation diagram for designing efficient dielectric microwave absorbers. The optimal microwave absorption parameters are screened based on parameter sweeping, and various graphene-based composites are fabricated to achieve strong microwave absorption (57 dB with full-band qualified absorption in X band). Although significant progress on carbon-based absorbers has been achieved, there are still several challenges to be addressed. Carbon-based absorbers require a large material thickness to achieve low-frequency microwave absorption, which greatly limits the development of coated carbon-based absorbers. In addition, it is difficult to realize broadband microwave absorption in carbon-based absorbers due to the absence of effective magnetic-dielectric synergy [52–55]. In order to address these challenges, these aspects should be concerned in future research. Firstly, the EM response of carbon materials with the uniform structure from nanometer size to micrometer size should be investigated to obtain the quantitative response of polarizability and relaxation time to microstructure, which can help the researchers to construct the desired

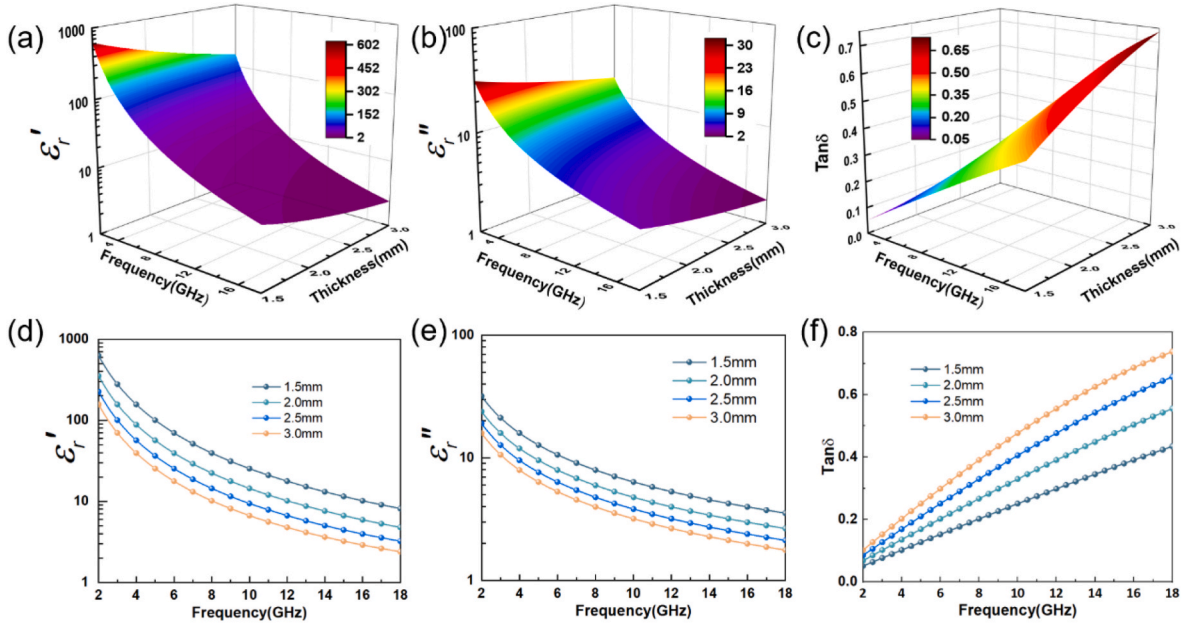



**Fig. 4.** 3D representations of (a) $\varepsilon_r^{'}$, (b) $\varepsilon_r^{''}$ and (c) Tanδ, frequency dependences of (d) $\varepsilon_r^{'}$, (e) $\varepsilon_r^{''}$ and (f) Tanδ with different thicknesses. (A colour version of this figure can be viewed online.)

**Table 1**
The complex permittivity of non-magnetic materials for perfect absorber with various thicknesses.

| Thickness (mm) | 4 GHz | 6 GHz | 8 GHz | 10 GHz | 12 GHz | 14 GHz | 16 GHz |
|---|---|---|---|---|---|---|---|
| 1.5 | 156.6–15.9i | 69.8–10.6i | 39.5–8.0i | 25.4–6.3i | 17.8-5.3i | 13.2–4.5i | 10.2–4.0i |
| 2.0 | 88.3–11.9i | 39.5–8.0i | 22.4–6.0i | 14.5-4.8i | 10.2–4.0i | 7.6-3.4i | 5.9-3.0i |
| 2.5 | 56.7–9.5i | 25.4–6.4i | 14.5-4.8i | 9.4-3.8i | 6.6-3.2i | 5.0–2.7i | 3.9-2.3i |
| 3.0 | 39.5–8.0i | 17.8-5.3i | 10.2–4.0i | 6.7-3.2i | 4.7-2.6i | 3.6-2.3i | 2.8-1.9i |

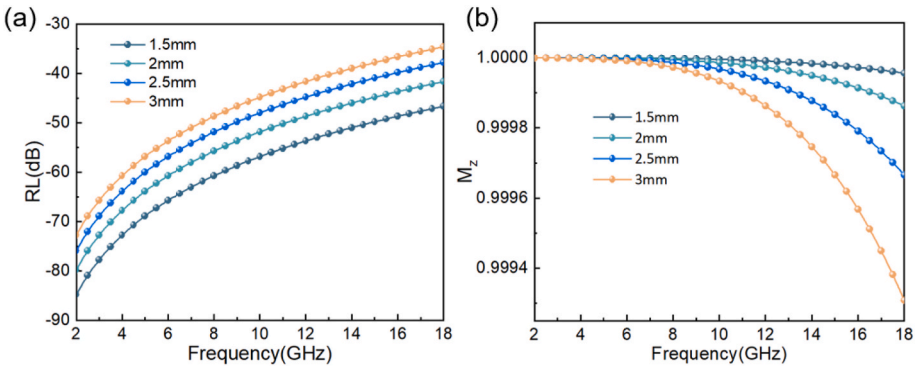



**Fig. 5.** Frequency dependences of (a) *RL* and (b) $M_z$ with different thicknesses. (A colour version of this figure can be viewed online.)

dielectric dispersion relation. Secondly, considering that the high-impedance surface formed by the resistive film can generate low-frequency resonance, multilayered carbon-based resistive films should be developed to solve the low-frequency microwave absorption challenge. Finally, discontinuously-distributed media in plane can produce unique electromagnetic response different from traditional dielectric materials, and then it is possible to obtain the dielectric dispersion relation with the coexistence of multiple resonances.

## 4. Summary and outlook

So far, numerous efforts have been devoted to developing advanced EMW absorbing materials for broadband and strong attenuation, and remarkable progress has been achieved. However, huge challenges remain in the practical. In the perspective, the progress of high-performance broadband absorbing materials has been briefly summarized. Impedance matching theory and quarter-wave resonance mechanism are dissected in detail from a unique perspective. On this basis, we construct an ideal dielectric dispersion model for the design of broadband microwave or light absorbers. Of course, the ideal dispersion permittivity for perfect absorption remains a challenge, due to the fact that almost all materials based on current major relaxation mechanisms such as Debye relaxation and Drude relaxation cannot satisfy this dispersion relationship. Significantly, the continuous resonance of multiple frequency bands is favorable to realizing broadband absorption. Therefore, as-proposed model can be used to construct multiple resonances co-existing in different frequency bands to achieve broadband absorption.

Toward the future, the following viewpoints and inquiries need to be considered for further development of high-performance broadband EMW absorbing materials. (1) Since theory can be a powerful tool for accelerating the discovery of new EMW absorbing materials, researchers should focus on finding new materials or constructing unique structures that can satisfy the dielectric dispersion relation. Under this premise, the effective absorption bandwidth can be expanded by constructing multi-resonance coordination in the desired frequency band. It can be prospected that multiple resonance coexistence will be an attractive target in the future. For increased effectiveness, research should shift from solely looking for new materials to understanding the fundamentals between relaxation and structure or particle size in order to successfully solve the broadband absorption problem. (2) For a microwave absorbing thin layer with a metal backplate, quarter-wavelength resonance or input impedance matching coefficient can be used to design strong attenuation absorbing materials. When the input impedance matching coefficient is greater than 0.82 and 0.98, the reflection loss of the corresponding frequency is less than −10dB and −20dB, respectively. A good input impedance matching will be achieved by adjusting the EM parameters of the material for the purpose of enhancing the absorption strength of the microwave absorber. For microwave absorbing materials with thickness significantly greater than the incident wavelength or particularly high-frequency EMW such as terahertz, however, the quarter-wavelength resonance and input impedance matching coefficient will fail in this case. To address the impedance mismatch problem in EMW absorbing materials with electrically large size, research efforts should be redirected from input impedance matching to both wave impedance matching coefficient $M_\eta$ and attenuation constant. Under this premise, to obtain a good absorbing performance, it is required to simultaneously increase $M_\eta$ and α by adjusting the electromagnetic parameters of the materials. The information provided from this perspective, we believe, will inspire researchers to make the development of wave-absorbing materials more efficient and meet the needs of the information technology fields.

## CRediT authorship contribution statement

**Zhi-Ling Hou:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Xuesong Gao:** Writing – original draft, Investigation, Data curation. **Junying Zhang:** Writing – review & editing, Visualization, Validation. **Guangsheng Wang:** Writing – review & editing, Visualization, Validation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this perspective.

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